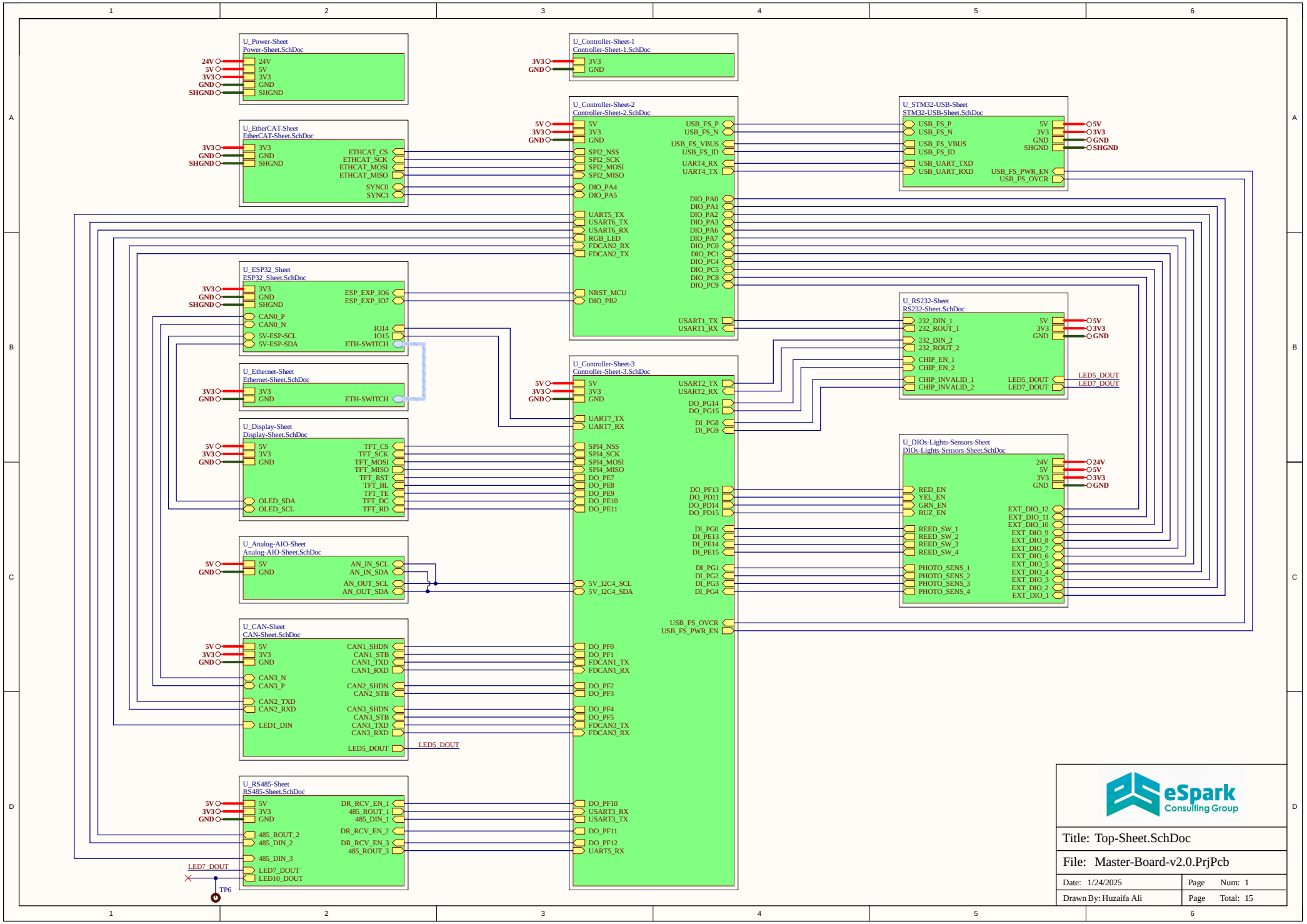
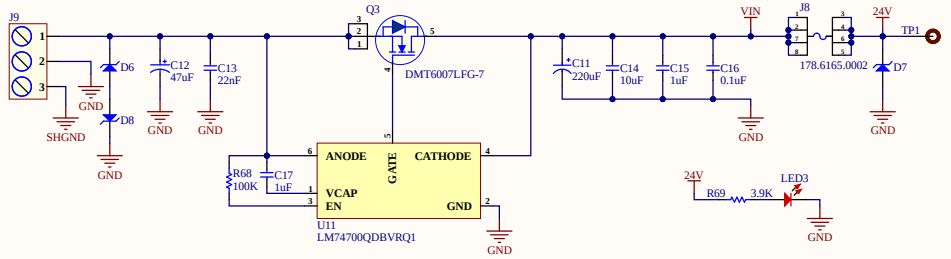
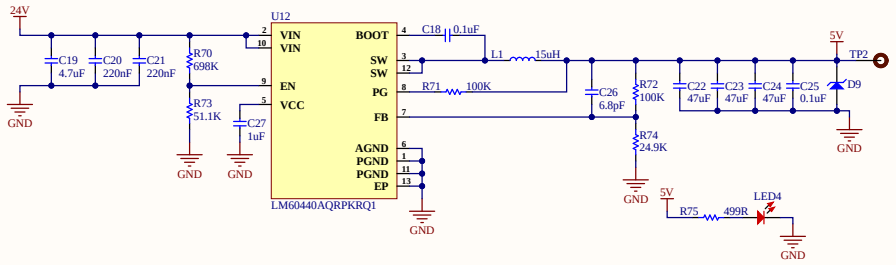

U_Power-Sheet
Power-Sheet.SchDoc
24V
5V
3V3
GND
SHGND
U_Controller-Sheet-1
Controller-Sheet-1.SchDoc
3V3
GND
U_EtherCAT-Sheet
EtherCAT-Sheet.SchDoc
ETHCAT_CS
ETHCAT_SCK
ETHCAT_MOSI
ETHCAT_MISO
SYNC0
SYNC1
U_Controller-Sheet-2
Controller-Sheet-2.SchDoc
USB_FS_P
USB_FS_N
USB_FS_VBUS
USB_FS_ID
UART4_RX
UART4_TX
SPI2_NSS
SPI2_SCK
SPI2_MOSI
SPI2_MISO
DIO_PA4
DIO_PA5
UART5_TX
USART6_TX
USART6_RX
RGB_LED
FDCAN2_RX
FDCAN2_TX
NRST_MCU
DIO_PB2
DIO_PA0
DIO_PA1
DIO_PA2
DIO_PA3
DIO_PA6
DIO_PA7
DIO_PC0
DIO_PC1
DIO_PC4
DIO_PC5
DIO_PC8
DIO_PC9
USART1_TX
USART1_RX
U_STM32-USB-Sheet
STM32-USB-Sheet.SchDoc
USB_FS_P
USB_FS_N
USB_FS_VBUS
USB_FS_ID
USB_UART_TXD
USB_UART_RXD
USB_FS_PWR_EN
USB_FS_OVCR
U_ESP32_Sheet
ESP32_Sheet.SchDoc
ESP_EXP_IO6
ESP_EXP_IO7
CAN0_P
CAN0_N
5V-ESP-SCL
5V-ESP-SDA
IO14
IO15
ETH-SWITCH
U_Ethernet-Sheet
Ethernet-Sheet.SchDoc
ETH-SWITCH
U_Display-Sheet
Display-Sheet.SchDoc
TFT_CS
TFT_SCK
TFT_MOSI
TFT_MISO
TFT_RST
TFT_BL
TFT_TE
TFT_DC
TFT_RD
OLED_SDA
OLED_SCL
U_Controller-Sheet-3
Controller-Sheet-3.SchDoc
USART2_TX
USART2_RX
DO_PG14
DO_PG15
DI_PG8
DI_PG9
UART7_TX
UART7_RX
SPI4_NSS
SPI4_SCK
SPI4_MOSI
SPI4_MISO
DO_PE7
DO_PE8
DO_PE9
DO_PE10
DO_PE11
DO_PF13
DO_PD11
DO_PD14
DO_PD15
DI_PG0
DI_PE13
DI_PE14
DI_PE15
DI_PG1
DI_PG2
DI_PG3
DI_PG4
5V_I2C4_SCL
5V_I2C4_SDA
USB_FS_OVCR
USB_FS_PWR_EN
U_RS232-Sheet
RS232-Sheet.SchDoc
232_DIN_1
232_ROUT_1
232_DIN_2
232_ROUT_2
CHIP_EN_1
CHIP_EN_2
CHIP_INVALID_1
CHIP_INVALID_2
LED5_DOUT
LED7_DOUT
U_DIOs-Lights-Sensors-Sheet
DIOs-Lights-Sensors-Sheet.SchDoc
RED_EN
YEL_EN
GRN_EN
BUZ_EN
REED_SW_1
REED_SW_2
REED_SW_3
REED_SW_4
PHOTO_SENS_1
PHOTO_SENS_2
PHOTO_SENS_3
PHOTO_SENS_4
EXT_DIO_12
EXT_DIO_11
EXT_DIO_10
EXT_DIO_9
EXT_DIO_8
EXT_DIO_7
EXT_DIO_6
EXT_DIO_5
EXT_DIO_4
EXT_DIO_3
EXT_DIO_2
EXT_DIO_1
U_Analog-AIO-Sheet
Analog-AIO-Sheet.SchDoc
AN_IN_SCL
AN_IN_SDA
AN_OUT_SCL
AN_OUT_SDA
U_CAN-Sheet
CAN-Sheet.SchDoc
CAN1_SHDN
CAN1_STB
CAN1_TXD
CAN1_RXD
CAN2_SHDN
CAN2_STB
CAN3_SHDN
CAN3_STB
CAN3_TXD
CAN3_RXD
CAN3_N
CAN3_P
CAN2_TXD
CAN2_RXD
LED1_DIN
LED5_DOUT
DO_PF0
DO_PF1
FDCAN1_TX
FDCAN1_RX
DO_PF2
DO_PF3
DO_PF4
DO_PF5
FDCAN3_TX
FDCAN3_RX
U_RS485-Sheet
RS485-Sheet.SchDoc
DR_RCV_EN_1
485_ROUT_1
485_DIN_1
DR_RCV_EN_2
DR_RCV_EN_3
485_ROUT_3
485_ROUT_2
485_DIN_2
485_DIN_3
LED7_DOUT
LED10_DOUT
TP6
DO_PF10
USART3_RX
USART3_TX
DO_PF11
DO_PF12
UART5_RX
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Consulting Group
Title: Top-Sheet.SchDoc
File: Master-Board-v2.0.PrjPcb
Date: 1/24/2025
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Drawn By: Huzaifa Ali
Page Total: 15

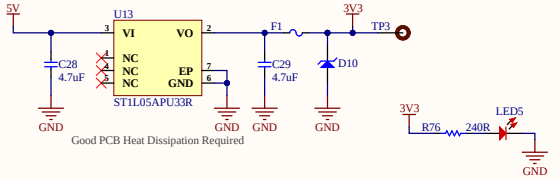
24V INPUT

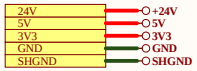


5V BUCK CONVERTER



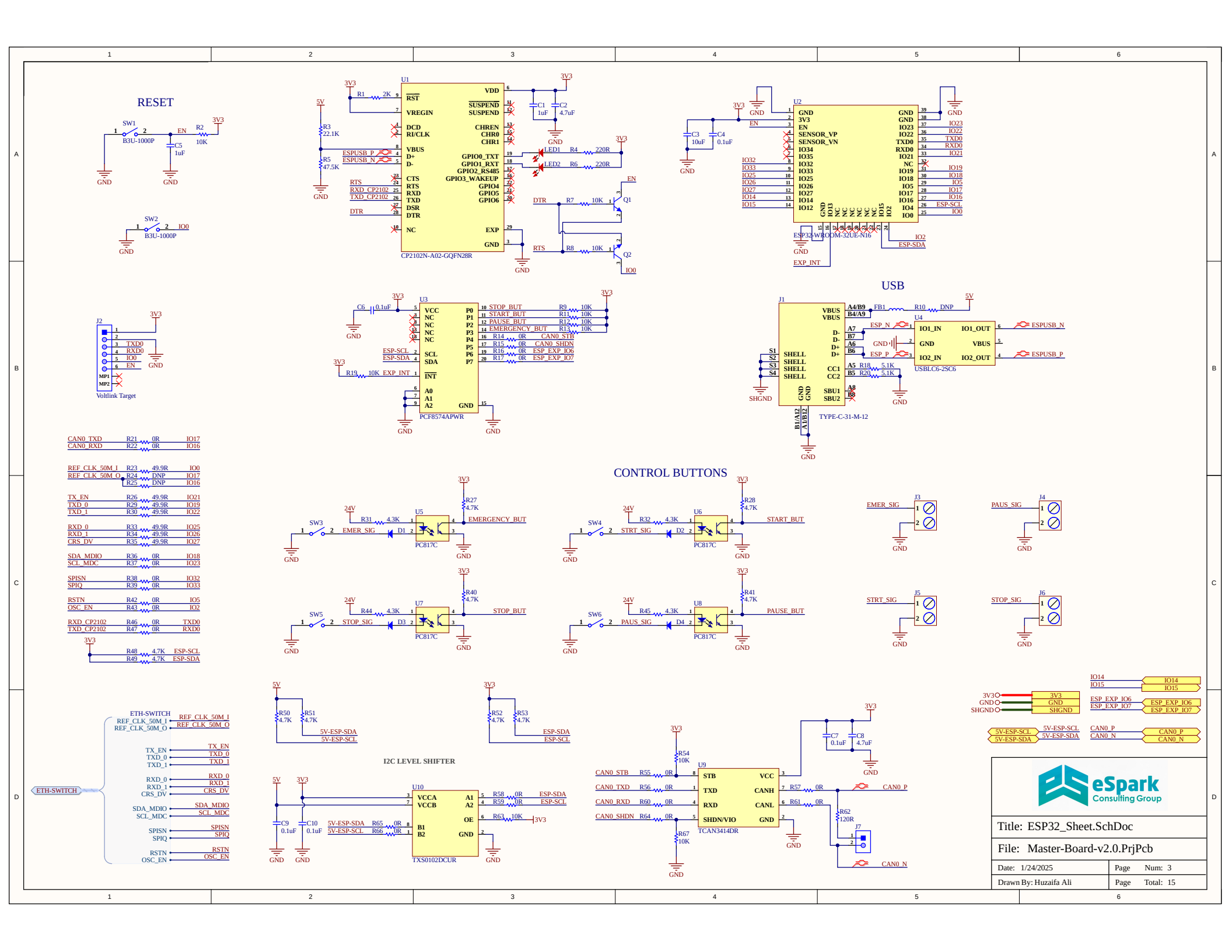
3.3V LINEAR REGULATOR



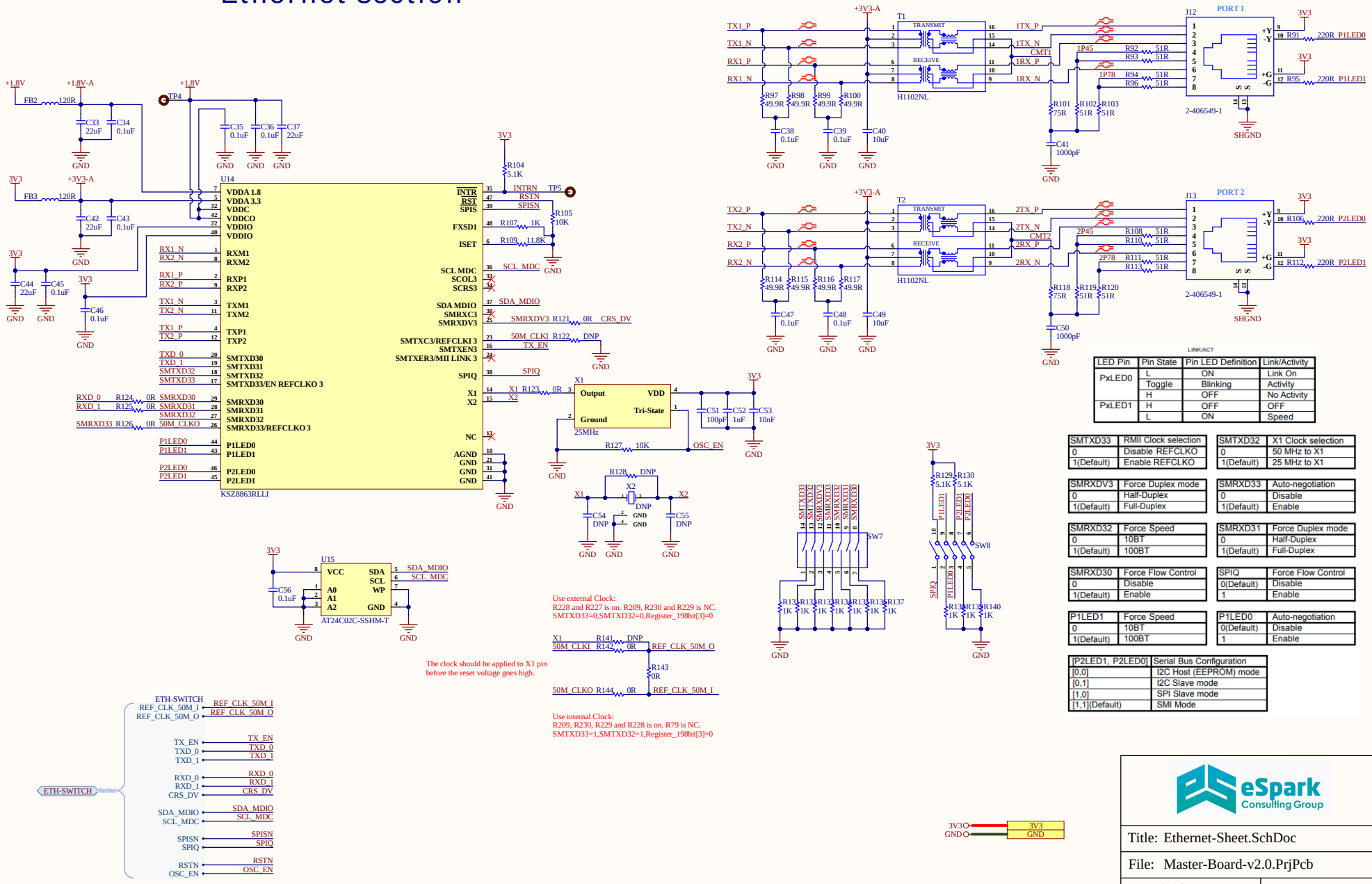


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Title: Power-Sheet.SchDoc	
File: Master-Board-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 2
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RESET
SW1
B3U-1000P
EN
R2
10K
3V3
C5
1uF
GND
SW2
IO0
U1
CP2102N-A02-GQFN28R
RST
VDD
VREGIN
SUSPEND
SUSPEND
DCD
RI/CLK
CHREN
CHR0
CHR1
VBUS
D+
D-
GPIO0_TXT
GPIO1_RXT
GPIO2_RS485
GPIO3_WAKEUP
CTS
RTS
RXD
TXD
DSR
DTR
GPIO4
GPIO5
GPIO6
NC
EXP
R1
2K
R3
22.1K
R5
47.5K
ESPUSB_P
ESPUSB_N
RTS
RXD_CP2102
TXD_CP2102
DTR
C1
1uF
C2
4.7uF
LED1
LED2
R4
220R
R6
220R
R7
10K
R8
10K
Q1
Q2
U2
ESP32-WROOM-32UE-N16
GND
3V3
EN
SENSOR_VP
SENSOR_VN
IO34
IO35
IO32
IO33
IO25
IO26
IO27
IO14
IO12
IO13
IO15
IO2
IO0
IO4
IO16
IO17
IO5
IO18
IO19
IO21
RXD0
TXD0
IO22
IO23
C3
10uF
C4
0.1uF
EXP_INT
ESP-SCL
ESP-SDA
J2
Voltlink Target
TXD0
RXD0
IO0
EN
MP1
MP2
U3
PCF8574APWR
VCC
NC
P0
P1
P2
P3
P4
P5
P6
P7
SCL
SDA
INT
A0
A1
A2
C6
0.1uF
R19
10K
STOP_BUT
START_BUT
PAUSE_BUT
EMERGENCY_BUT
R9
R11
R12
R13
R14
R15
R16
R17
0R
CAN0_STB
CAN0_SHDN
ESP_EXP_IO6
ESP_EXP_IO7
USB
J1
TYPE-C-31-M-12
VBUS
A4/B9
B4/A9
A7
B7
A6
B6
SHELL
CC1
CC2
SBU1
SBU2
A5
B5
A8
B8
B1/A12
A1/B12
SHGND
FB1
R10
DNP
R18
R20
5.1K
U4
USBLC6-2SC6
IO1_IN
IO1_OUT
IO2_IN
IO2_OUT
ESPUSB_N
ESPUSB_P
ESP_N
ESP_P
CONTROL BUTTONS
SW3
SW4
SW5
SW6
EMER_SIG
STRT_SIG
STOP_SIG
PAUS_SIG
R31
R32
R44
R45
4.3K
24V
D1
D2
D3
D4
PC817C
U5
U6
U7
U8
R27
R28
R40
R41
4.7K
J3
J4
J5
J6
CAN0_TXD
CAN0_RXD
R21
R22
IO17
IO16
REF_CLK_50M_I
REF_CLK_50M_O
R23
R24
R25
49.9R
IO0
IO17
IO16
TX_EN
TXD_0
TXD_1
R26
R29
R30
IO21
IO19
IO22
RXD_0
RXD_1
CRS_DV
R33
R34
R35
IO25
IO26
IO27
SDA_MDIO
SCL_MDC
R36
R37
IO18
IO23
SPISN
SPIQ
R38
R39
IO32
IO33
RSTN
OSC_EN
R42
R43
IO5
IO2
RXD_CP2102
TXD_CP2102
R46
R47
TXD0
RXD0
R48
R49
ESP-SCL
ESP-SDA
ETH-SWITCH
REF_CLK_50M_I
REF_CLK_50M_O
TX_EN
TXD_0
TXD_1
RXD_0
RXD_1
CRS_DV
SDA_MDIO
SCL_MDC
SPISN
SPIQ
RSTN
OSC_EN
I2C LEVEL SHIFTER
U10
TXS0102DCUR
VCCA
VCCB
A1
A2
B1
B2
OE
R50
R51
R52
R53
R58
R59
R63
R65
R66
5V-ESP-SDA
5V-ESP-SCL
C9
C10
U9
TCAN3414DR
STB
TXD
RXD
SHDN/VIO
VCC
CANH
CANL
R54
R55
R56
R60
R64
R67
R57
R61
R62
120R
C7
C8
4.7uF
J7
CAN0_P
CAN0_N
IO14
IO15
ESP_EXP_IO6
ESP_EXP_IO7
3V3
GND
SHGND
5V-ESP-SCL
5V-ESP-SDA
eSpark
Consulting Group
Title: ESP32_Sheet.SchDoc
File: Master-Board-v2.0.PrjPcb
Date: 1/24/2025
Page Num: 3
Drawn By: Huzaifa Ali
Page Total: 15

Ethernet Section



LED Pin	Pin State	Pin LED Definition	Link/Activity
PxLED0	L	ON	Link On
	Toggle	Blinking	Activity
	H	OFF	No Activity
PxLED1	H	OFF	OFF
	L	ON	Speed

LINK/ACT

SMTXD33	RMII Clock selection
0	Disable REFCLKO
1(Default)	Enable REFCLKO

SMTXD32	X1 Clock selection
0	50 MHz to X1
1(Default)	25 MHz to X1

SMRXDV3	Force Duplex mode
0	Half-Duplex
1(Default)	Full-Duplex

SMRXD33	Auto-negotiation
0	Disable
1(Default)	Enable

SMRXD32	Force Speed
0	10BT
1(Default)	100BT

SMRXD31	Force Duplex mode
0	Half-Duplex
1(Default)	Full-Duplex

SMRXD30	Force Flow Control
0	Disable
1(Default)	Enable

SPIQ	Force Flow Control
0(Default)	Disable
1	Enable

P1LED1	Force Speed
0	10BT
1(Default)	100BT

P1LED0	Auto-negotiation
0(Default)	Disable
1	Enable

[P2LED1, P2LED0]	Serial Bus Configuration
[0,0]	I2C Host (EEPROM) mode
[0,1]	I2C Slave mode
[1,0]	SPI Slave mode
[1,1](Default)	SMI Mode

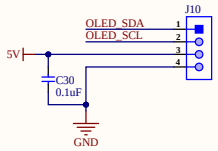
Use external Clock:
R228 and R227 is on. R209, R230 and R229 is NC.
SMTXD33=0,SMTXD32=0,Register_198bit[3]=0

The clock should be applied to X1 pin before the reset voltage goes high.

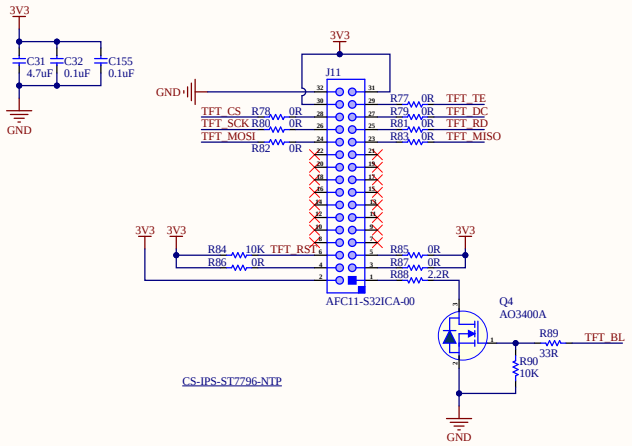
Use internal Clock:
R209, R230, R229 and R228 is on. R79 is NC.
SMTXD33=1,SMTXD32=1,Register_198bit[3]=0

Title:	Ethernet-Sheet.SchDoc
File:	Master-Board-v2.0.PrjPcb
Date: 1/24/2025	Page Num: 4
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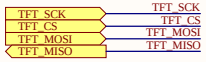
OLED DISPLAY

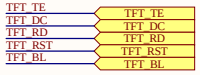


TFT DISPLAY







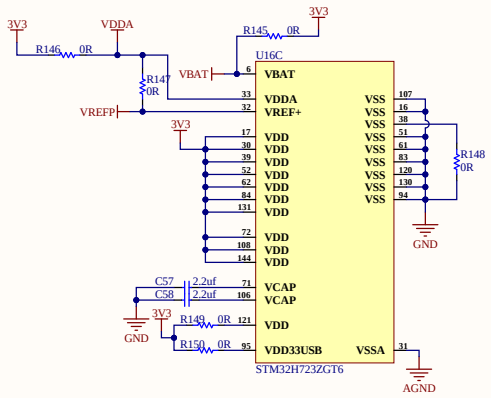


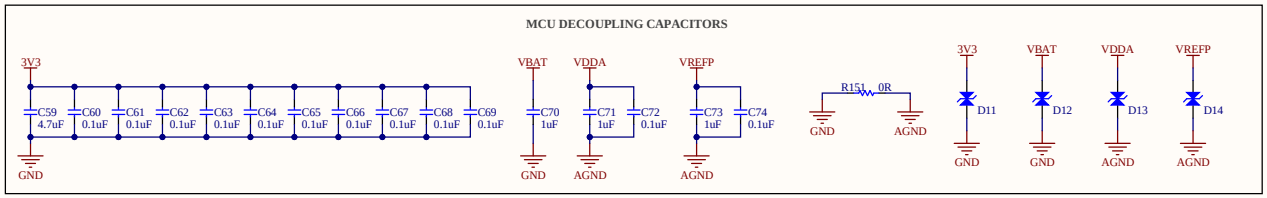


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Title: Display-Sheet.SchDoc	
File: Master-Board-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 5
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CONTROLLER POWER SHEET



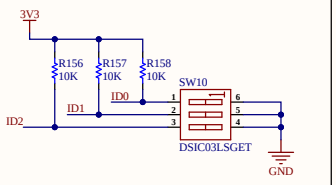




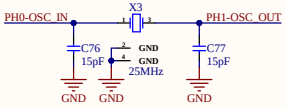
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File: Master-Board-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 15

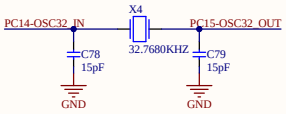
CONTROLLER IO SHEET 1

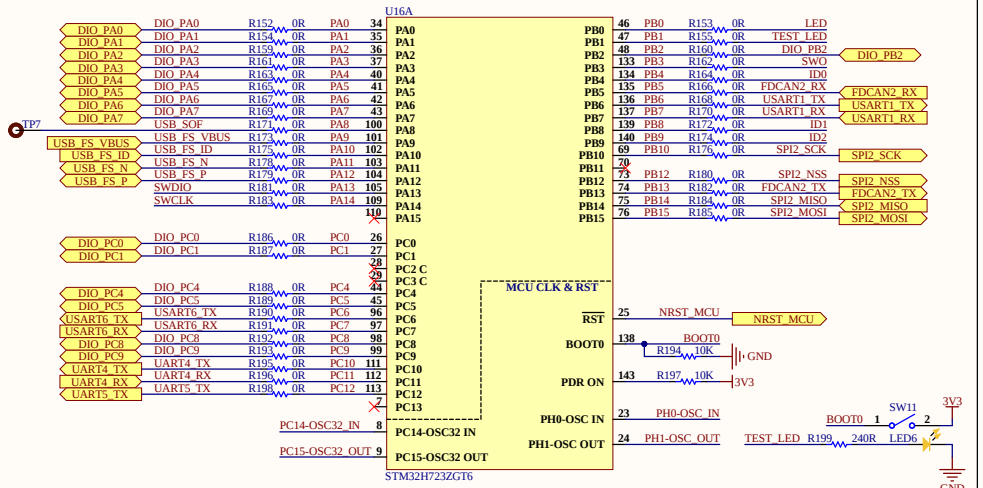


EXTERNAL HSE CLOCK

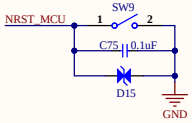


EXTERNAL LSE CLOCK

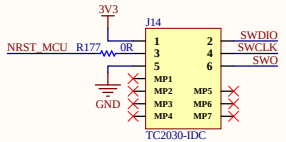


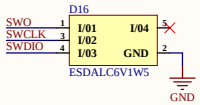


RESET FUNCTION

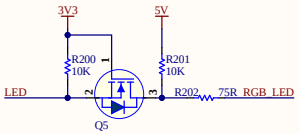


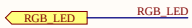
PRORAMMING TAG CONNECT





RGB LED LEVEL TRANSLATOR



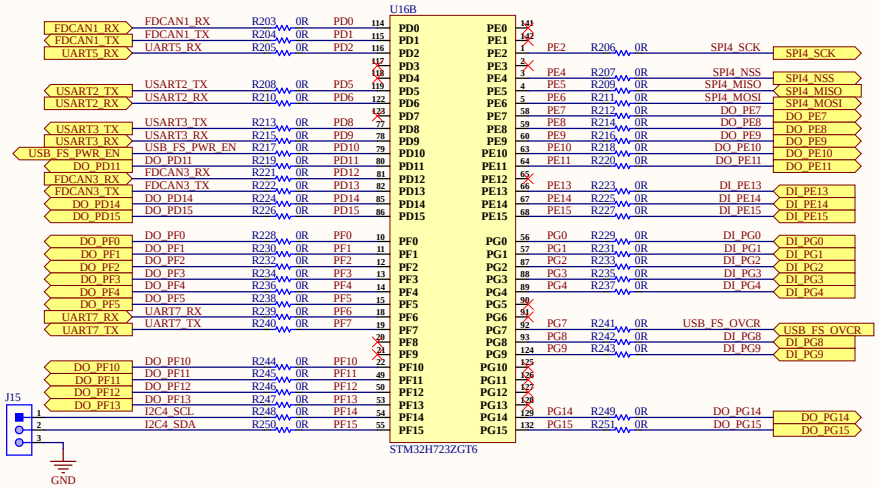




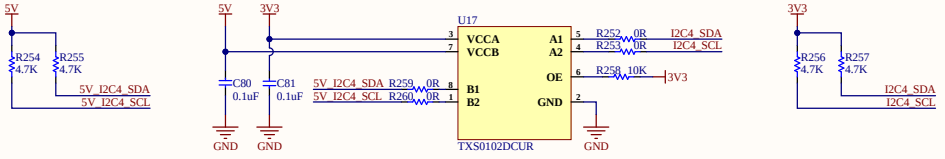
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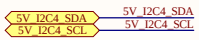
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File: Master-Board-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 15

CONTROLLER IO SHEET 2



I2C LEVEL SHIFTER



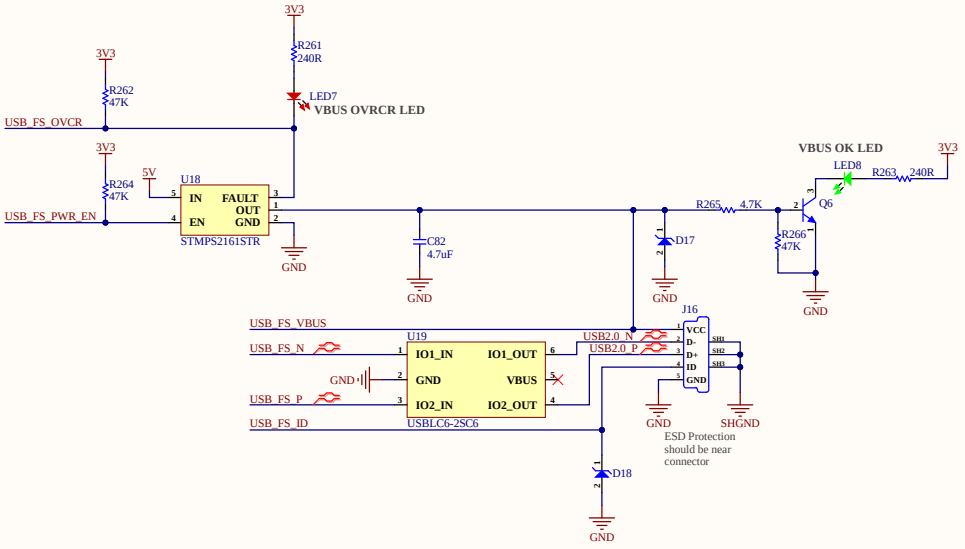




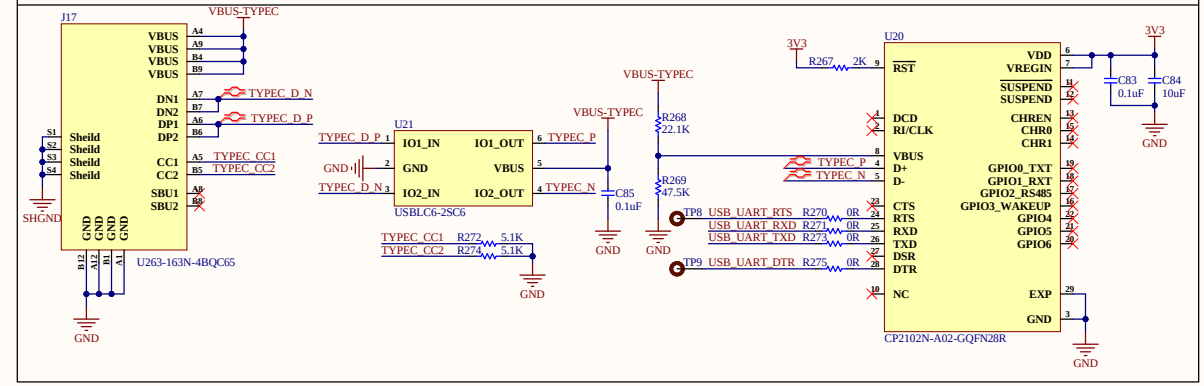
eSpark Consulting Group

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Date: 1/24/2025	Page Num: 8
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MICRO USB



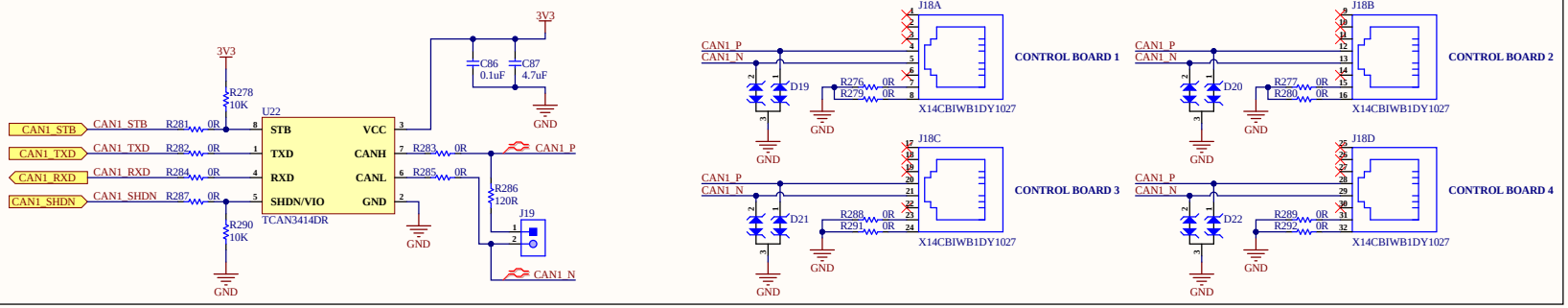
DEBUG TYPE-C



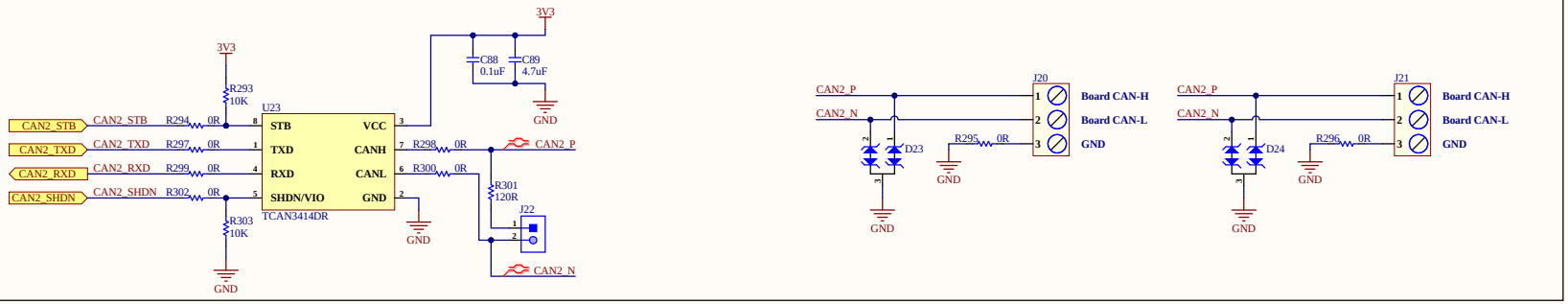


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Drawn By: Huzaifa Ali	Page Total: 15

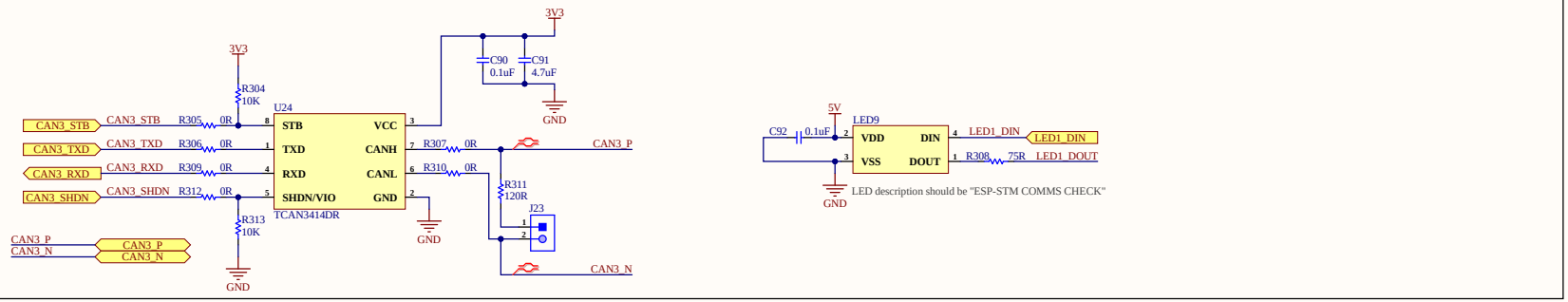
CAN 1



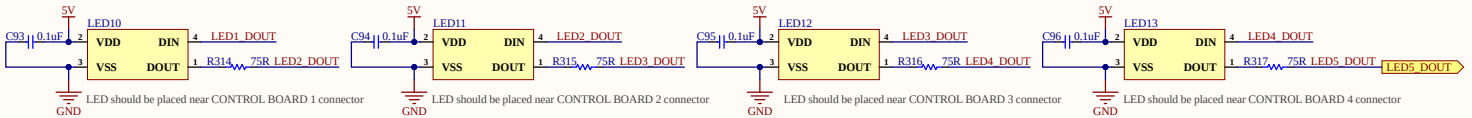
CAN 2



CAN 3



LEDs

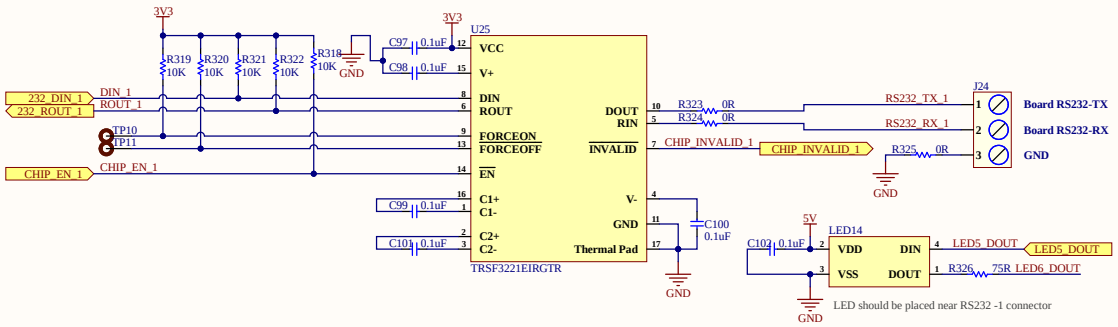


5V 5V
3V3 3V3
GND GND

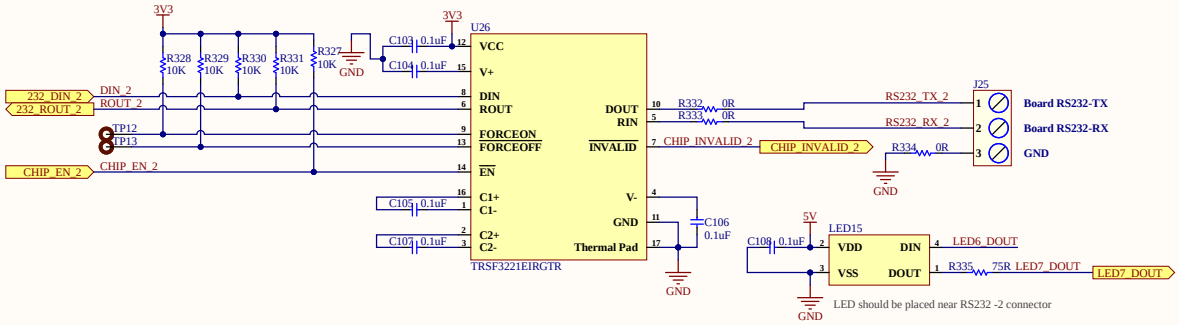
eSpark Consulting Group

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RS232 - 1



RS232 - 2

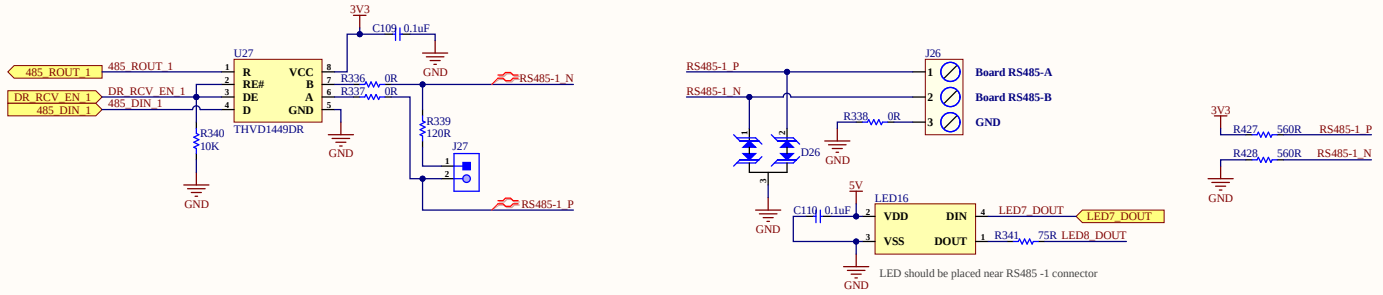




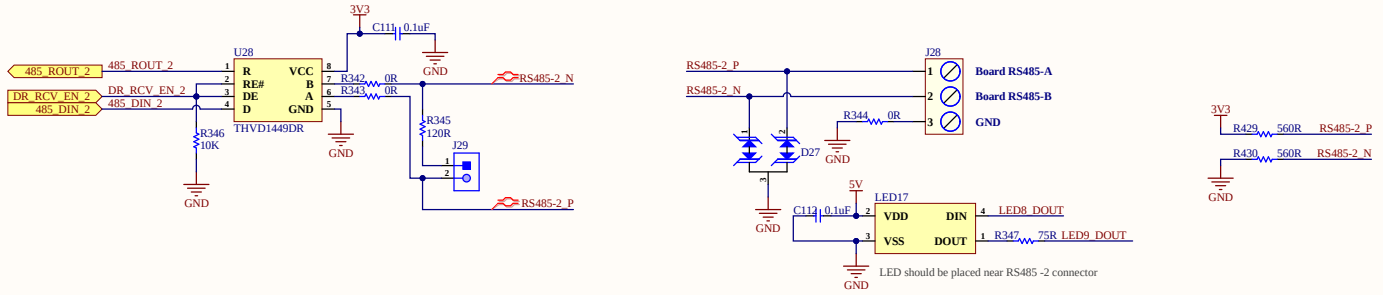
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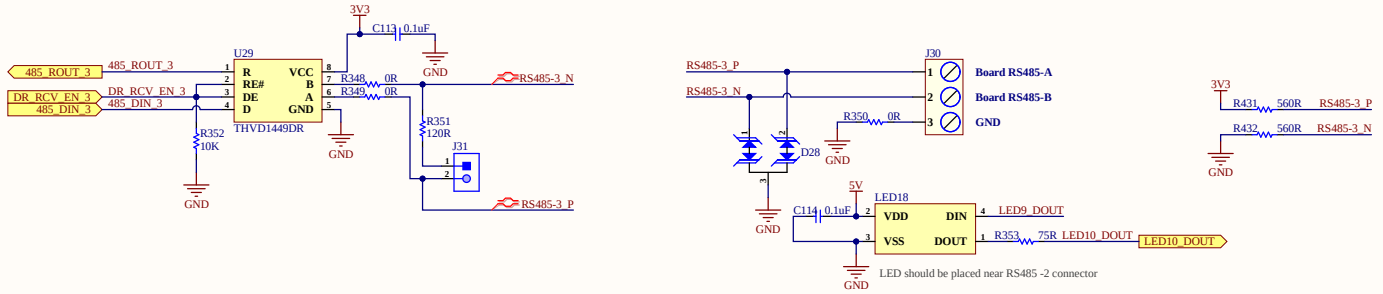
RS485 - 1

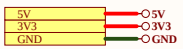


RS485 - 2



RS485 - 3

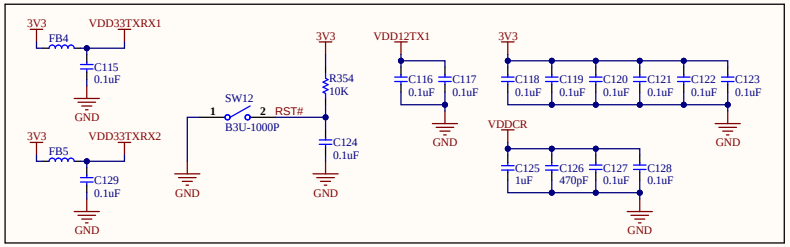


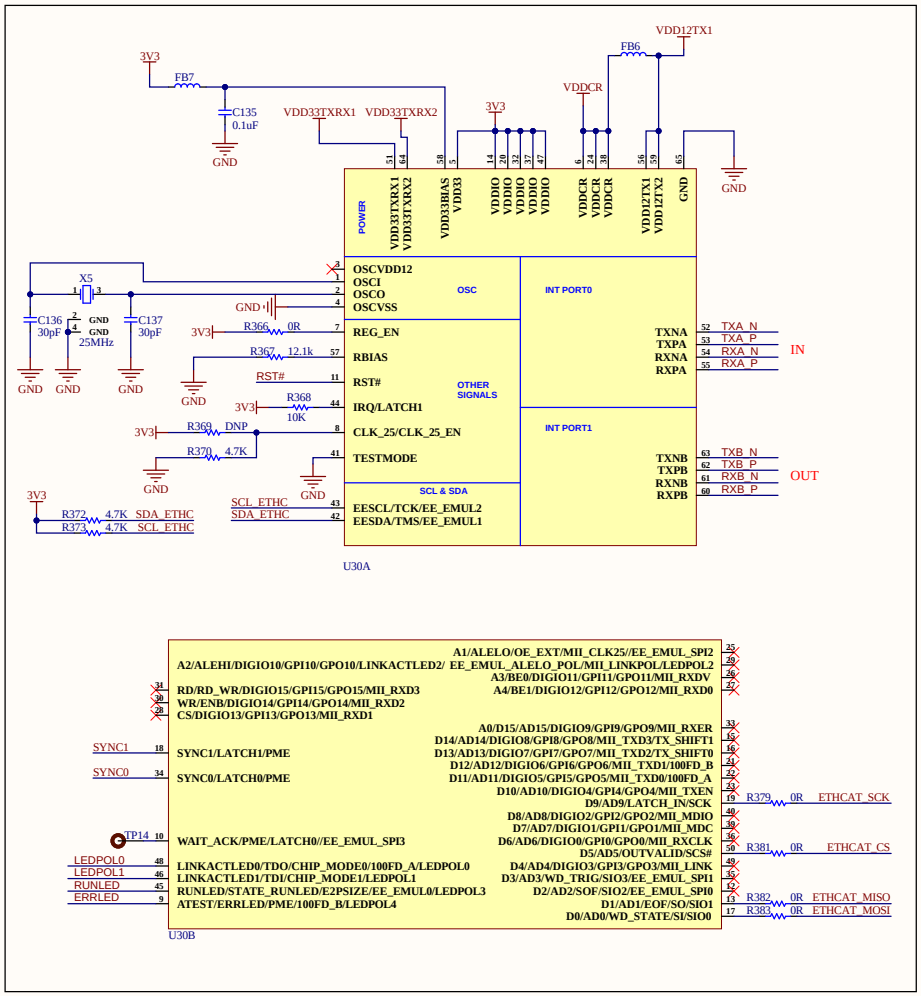


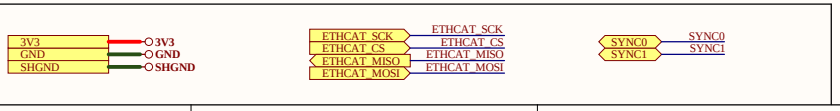
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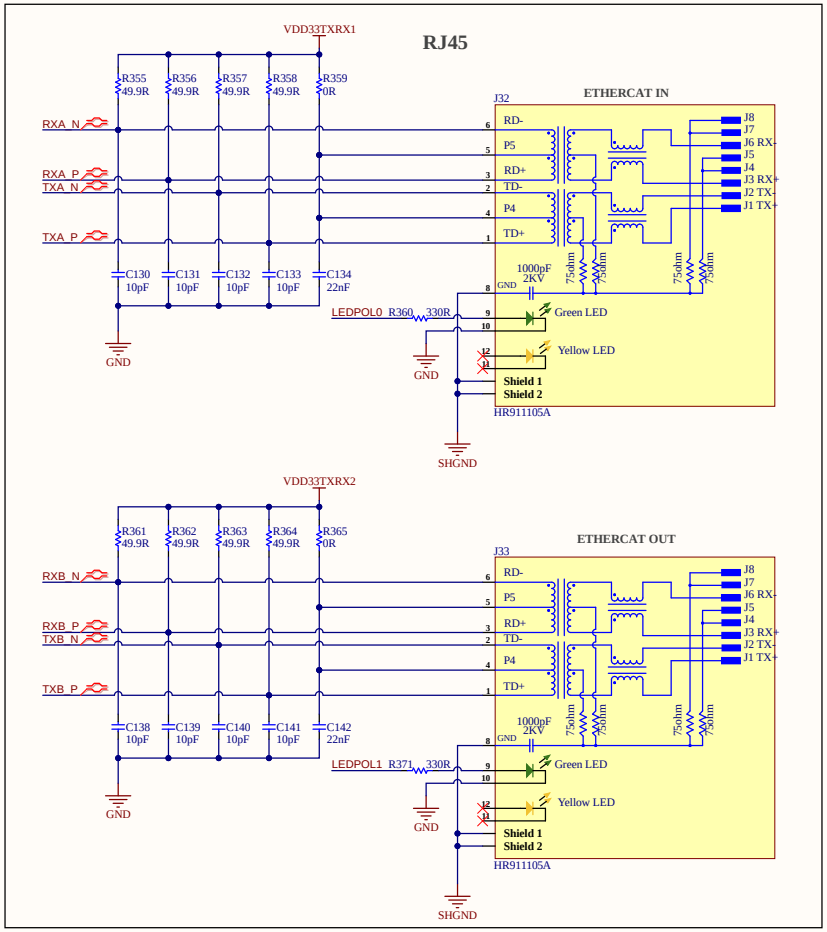
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Date: 1/29/2025	Page Num: 12
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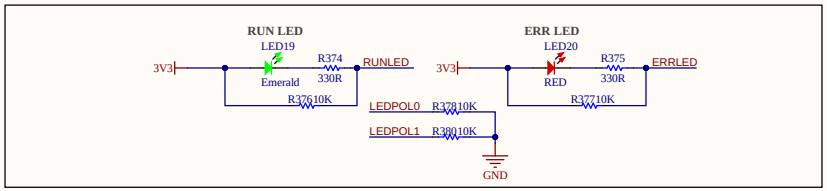
ETHERCAT

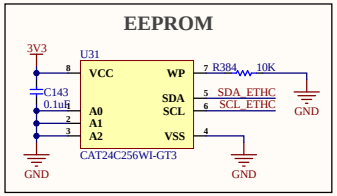








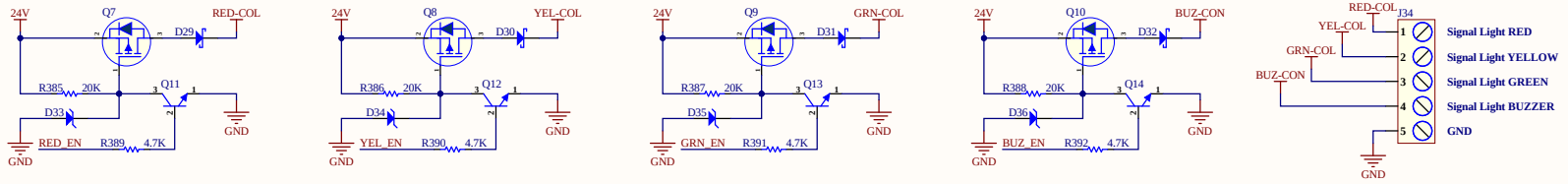




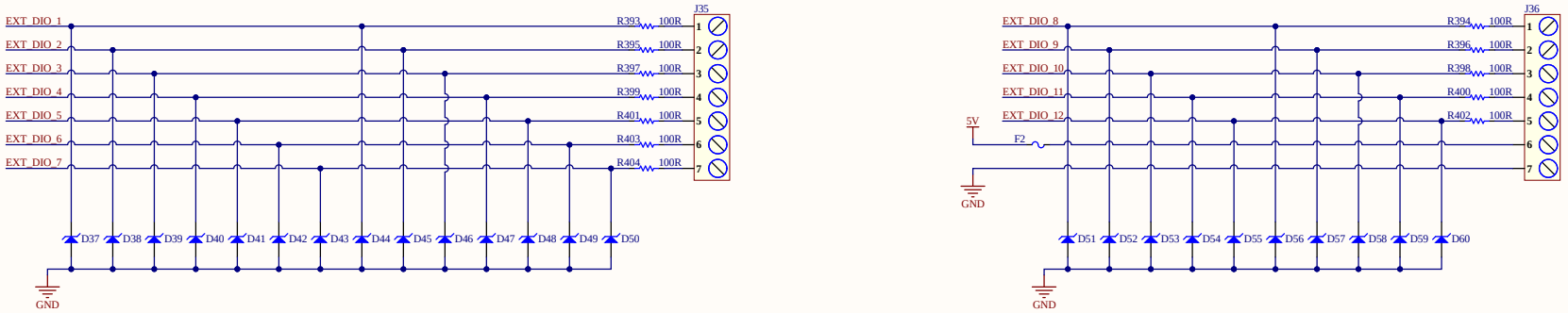
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Date: 1/24/2025	Page Num: 13
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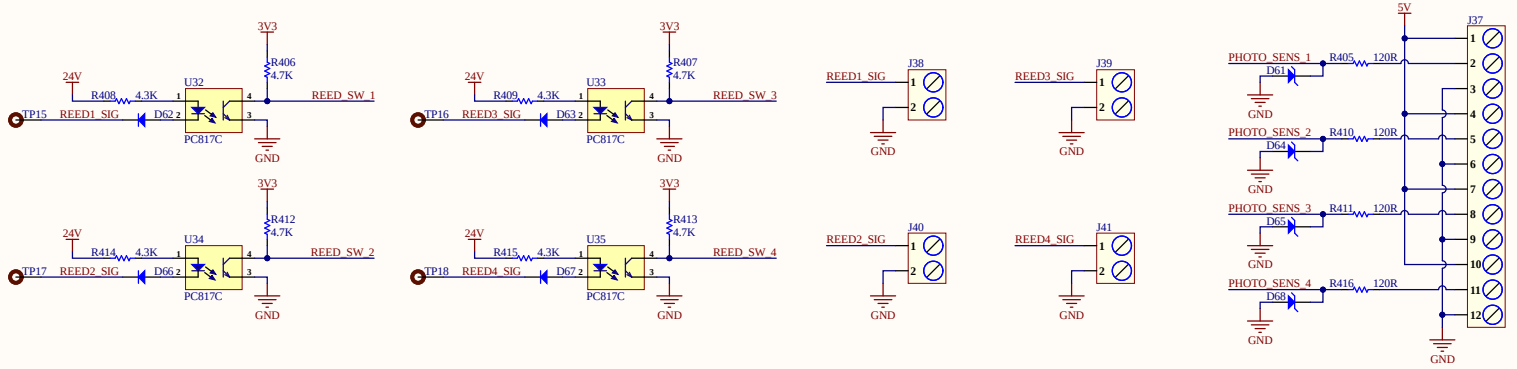
SIGNAL LIGHTS

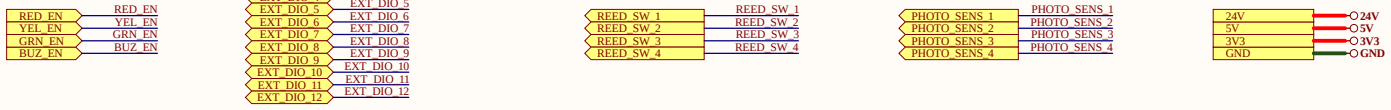


EXTRA DIOs



REED AND PHOTOELECTRIC SENSOR

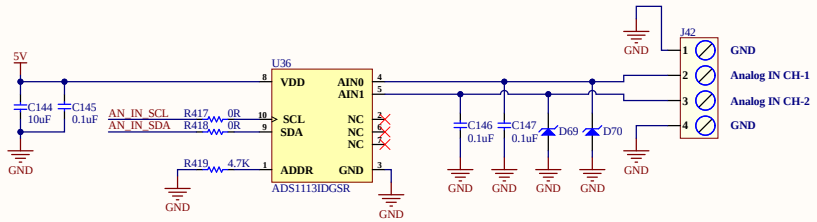




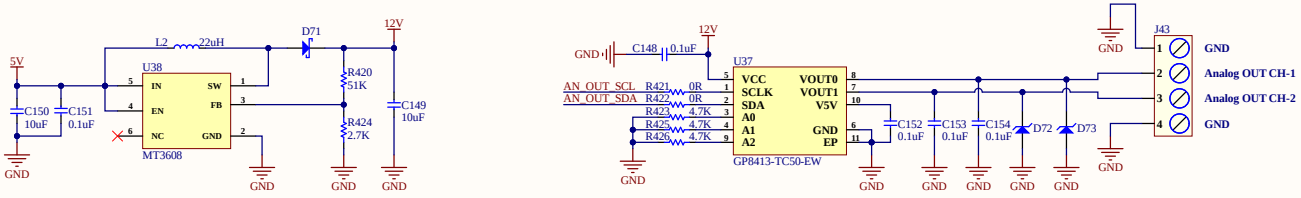
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Title: DIOs-Lights-Sensors-Sheet.SchDoc	
File: Master-Board-v2.0.PrjPcb	
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ANALOG INPUTS 0-5V



ANALOG OUTPUTS 0-5V



AN_IN_SCL — AN_IN_SCL
AN_IN_SDA — AN_IN_SDA

AN_OUT_SCL — AN_OUT_SCL
AN_OUT_SDA — AN_OUT_SDA

5V — 5V
GND — GND

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Title: Analog-AIO-Sheet.SchDoc	
File: Master-Board-v2.0.PrjPcb	
Date: 1/24/2025	Page Num: 15
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